STAT 1210 - Quiz 2 Solutions - Version 3

Part II: Probability and Distributions

Fall 2026

Question 1: Calibration Time for a Laboratory Instrument (4 marks)

1. The possible calibration times extend from 1 to 9 minutes, so the total width of the uniform distribution is

$$9 - 1 = 8 \text{ minutes}$$

The uniform density forms a rectangle with total area 1. If h is its height, then

$$8h = 1 \qquad \boxed{h = \frac{1}{8}}$$

2. The interval from 2.4 to 6.4 minutes has width

$$6.4 - 2.4 = 4.0 \text{ minutes}$$

The required probability is the area of the rectangle over this interval:

$$P(2.4 \leq T \leq 6.4) = (\text{width})(\text{height}) = (4.0)\left(\frac{1}{8}\right) = \boxed{0.500}$$

3. The two relevant intervals are disjoint. Their widths are

$$2.2 - 1 = 1.2 \text{ minutes} \qquad \text{and} \qquad 9 - 8.2 = 0.8 \text{ minutes}$$

Therefore

$$P(T < 2.2 \text{ or } T > 8.2) = (1.2)\left(\frac{1}{8}\right) + (0.8)\left(\frac{1}{8}\right) = \frac{2.0}{8} = \boxed{0.250}$$

Question 2: Parasitic Infection in Dogs (6 marks)

Let I denote parasitic infection and let D denote diarrhea.

1. The marginal and joint probabilities are

$$P(I) = \frac{86}{300} \approx \boxed{0.2867}$$

$$P(D) = \frac{88}{300} \approx \boxed{0.2933}$$

$$P(I \cap D) = \frac{50}{300} \approx \boxed{0.1667}$$

2. Conditioning on diarrhea restricts the denominator to the 88 dogs with diarrhea. Therefore

$$P(I \mid D) = \frac{P(I \cap D)}{P(D)} = \frac{50}{88} \approx \boxed{0.5682}$$

3. Conditioning on infection restricts the denominator to the 86 infected dogs. Therefore

$$P(D \mid I) = \frac{P(D \cap I)}{P(I)} = \frac{50}{86} \approx \boxed{0.5814}$$

The probabilities $P(I \mid D)$ and $P(D \mid I)$ both use the 50 dogs in the intersection, but they use different denominators. The first conditions on the 88 dogs with diarrhea whereas the second conditions on the 86 dogs with an infection.

Question 3: Recovery Time After Veterinary Anaesthesia (6 marks)

1. The recovery time is approximately Normal with

$$\mu = 40 \text{ minutes} \qquad \text{and} \qquad \sigma = 4 \text{ minutes}$$

The standardized values of the two boundaries are

$$z_{35} = \frac{35-40}{4} = -1.25 \qquad \text{and} \qquad z_{47} = \frac{47-40}{4} = 1.75$$

Therefore

$$P(35 < R < 47) = P(z_{35} < Z < z_{47}) = P(-1.25 < Z < 1.75) = P(Z \leq 1.75) - P(Z \leq -1.25) = N(1.75) - N(-1.25)$$

The standard Normal table gives

$$P(Z \leq -1.25) = N(-1.25) = 1 - N(1.25) = 1 - 0.8944 = 0.1056 \qquad \text{and} \qquad P(Z \leq 1.75) = 0.9599$$

The probability between the two boundaries is

$$P(35 < R < 47) = N(1.75) - N(-1.25) = 0.9599 - 0.1056 = \boxed{0.8543}$$

2. From the standardization and table values calculated above

$$P(R < 35) = P(Z < -1.25) = N(-1.25) = 1 - N(1.25) = 1 - 0.8944 = \boxed{0.1056}$$

For the upper-tail probability

$$P(R > 47) = P(Z > 1.75) = 1 - P(Z \leq 1.75) = 1 - N(1.75) = 1 - 0.9599 = \boxed{0.0401}$$

3. The expected number of recovery times between 35 and 47 minutes is

$$350(0.8543) = \boxed{299.005}$$

Approximately 299 of the 350 recovery times would be expected to fall within this interval.

Question 4: Calibration Failures in Radiation Sensors (4 marks)

1. There are 8 independent trials, each sensor is classified as either failing or not failing and the probability of failure is 0.10 for every sensor. Therefore

$$X \sim \text{Bin}(8, 0.10)$$

The phrase "no more than one sensor fails" means

$$\boxed{P(X \leq 1)}$$

2. The probability that no sensors fail is

$$P(X = 0) = \binom{8}{0}(0.10)^0(0.90)^8 = (0.90)^8 \approx \boxed{0.430}$$

3. The probability that exactly one sensor fails is

$$P(X = 1) = \binom{8}{1}(0.10)^1(0.90)^7 = 8(0.10)(0.90)^7 \approx \boxed{0.383}$$

4. The events $X = 0$ and $X = 1$ are disjoint, so

$$P(X \leq 1) = P(X = 0) + P(X = 1) \approx 0.430467 + 0.382638 \approx \boxed{0.813}$$

Under this model, there is approximately an 81.3% probability that no more than one of the 8 radiation sensors fails its calibration test.